THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 8 — MATHS · HARD LEVEL CHAPTER TEST SET 1

Chapter 4: **Quadrilaterals** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The angles of a quadrilateral are in the ratio 3:4:5:8. The difference between the largest and the smallest angle is: 1

(a) 198° **(b)** 102° **(c)** 144° **(d)** 90°

2. One angle of a parallelogram is 2 times its adjacent angle. The smaller angle is: 1

(a) 60° **(b)** 120° **(c)** 80° **(d)** 90°

3. In a rhombus ABCD, $\angle ACD = 25^\circ$. Then $\angle ABC$ is: 1

(a) 65° **(b)** 50° **(c)** 130° **(d)** 155°

4. The diagonals of a rectangle ABCD meet at O. If $\angle AOD = 100^\circ$, then $\angle OAB$ is: 1

(a) 40° **(b)** 50° **(c)** 65° **(d)** 100°

5. In an isosceles trapezium ABCD with AB $\parallel$ DC, $\angle A = (3x + 10)^\circ$ and $\angle D = (4x + 30)^\circ$. The measure of $\angle C$ is: 1

(a) 130° **(b)** 70° **(c)** 110° **(d)** 145°

6. The side of a rhombus is $(3x + 2)$ cm and its perimeter is 80 cm. The value of x is: 1

(a) 5 **(b)** 6 **(c)** 7 **(d)** 8

7. A quadrilateral has equal diagonals that do not bisect each other. It can be an: 1

(a) rhombus **(b)** kite
(c) parallelogram **(d)** isosceles trapezium

8. The angles of a quadrilateral are in the ratio 1:3:3:5. The difference between the largest and the smallest angle is: 1

(a) 180° **(b)** 150° **(c)** 132° **(d)** 120°

9. One angle of a parallelogram is 3 times its adjacent angle. The smaller angle is: 1

(a) 135° **(b)** 65° **(c)** 60° **(d)** 45°

10. In a rhombus ABCD, $\angle ACD = 32^\circ$. Then $\angle ABC$ is: 1

(a) 116° **(b)** 58° **(c)** 148° **(d)** 64°

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. In a parallelogram ABCD, $\angle A = (4x + 14)^\circ$ and $\angle C = (5x + 2)^\circ$. Find x and $\angle B$. 2

12. The angles of a quadrilateral are in the ratio 2:5:6:7. Find the largest angle and say whether it is acute, right or obtuse. 2

13. In a kite ABCD, AB = AD, CB = CD, $\angle A = 70^\circ$ and $\angle B = 100^\circ$. Find $\angle C$ and $\angle D$. 2

14. In a parallelogram ABCD, $\angle A = (3x + 30)^\circ$ and $\angle C = (5x + 10)^\circ$. Find x and $\angle B$. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. In a parallelogram ABCD with $\angle A = 50^\circ$, the bisectors of $\angle A$ and $\angle B$ meet at P. Find $\angle APB$. 3

16. The angles of a quadrilateral are x°, $(2x - 10)^\circ$, $(3x - 20)^\circ$ and $(4x - 30)^\circ$. Find all four angles. 3

17. One angle of a parallelogram is 2 times its adjacent angle. Find all the angles of the parallelogram. 3

OR

In a parallelogram ABCD with $\angle A = 70^\circ$, the bisectors of $\angle A$ and $\angle B$ meet at P. Find $\angle APB$. 3

18. The angles of a quadrilateral are x°, $(2x - 20)^\circ$, $(3x - 40)^\circ$ and $(4x - 60)^\circ$. Find all four angles. 3

OR

One angle of a parallelogram is 3 times its adjacent angle. Find all the angles of the parallelogram. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. In a quadrilateral ABCD, $\angle A : \angle B : \angle C : \angle D = 1 : 2 : 3 : 4$. Find the four angles. Show that AB $\parallel$ DC. 5

OR

In a parallelogram ABCD, AB $= (3x + 4)$ cm, CD $= (2x + 9)$ cm and AD = 12 cm. Find x, the lengths of AB and CD, and the perimeter. 5

20. In a quadrilateral ABCD, $\angle A : \angle B : \angle C : \angle D = 2 : 3 : 3 : 4$. Find the four angles. Show that AB $\parallel$ DC. 5

OR

In a parallelogram ABCD, AB $= (4x + 2)$ cm, CD $= (3x + 9)$ cm and AD = 15 cm. Find x, the lengths of AB and CD, and the perimeter. 5

END OF QUESTION PAPER — **Total: 40 Marks**

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